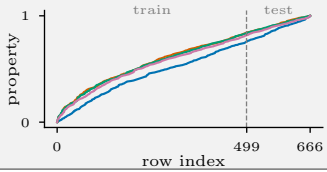
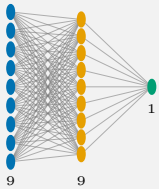
Master QCAR table

667 constructed rows: 9 elements (wt.%) → TS, PS, SS, HV



Baseline ANN workflow



random 70:15:15 division of all rows

$R \approx 0.9999$

on the random test subset

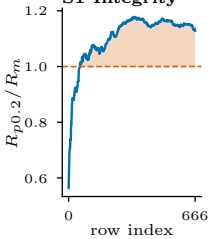
The question

Does $R \approx 0.9999$ show that the model can predict the properties of an *unseen* alloy?

Audit suite: six strategies that test the data, the split, the validation, the leakage, the interpretation, and the uncertainty

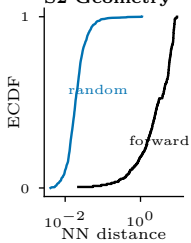
Audit suite and key findings

S1 Integrity



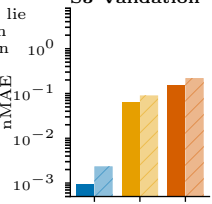
proof stress > tensile strength in 607/667 rows; source strengths copy a handbook temper sequence

S2 Geometry



forward test rows lie 160× farther from training rows than random holdouts

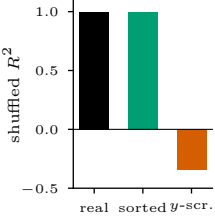
S3 Validation



$R^2 > 0.999$ (shuffled) becomes $R^2 < 0$ (forward); ANN R holds only for random splits

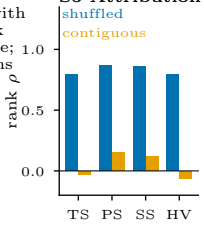
solid: Extra Trees
hatched: ANN replica

S4 Leakage



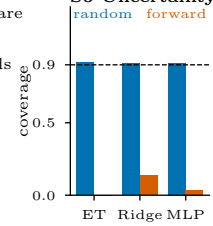
sorted surrogates with no composition link match the real table; row order alone wins

S5 Attribution



Cu, Mg, Mn, Zn are perfect proxies; rankings collapse in contiguous folds

S6 Uncertainty



90% intervals hold for random splits but cover 0–22% of forward rows

Conclusion: the near-perfect accuracy reflects row order and interpolation, not metallurgy.
Rebuild the data with provenance and grouped validation before interpretation or inverse alloy design.